

WiFi Sensing Security Overlay

A Proposal for Silent Nite Security

Prepared for: Luigi ? Silent Nite Security

Prepared by: Stefan (LAIS Research & Development)

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Status: Preliminary R&D Proposal

This document outlines a proposed research and development partnership between Stefan (LAIS) and Silent Nite Security. It is a preliminary proposal intended for discussion purposes only and does not constitute a binding agreement.

1. Executive Summary

This proposal presents a new type of security sensing layer that detects human presence, movement, and body position through walls using standard WiFi signals. The technology is complementary to existing alarm systems and CCTV installations ? it fills the gaps cameras cannot see.

The system uses small, low-cost ESP32 sensor nodes (R500-R800 each) deployed around a property. These nodes analyse how WiFi waves reflect off human bodies, then send the data to a central server where machine learning reconstructs the position and movement of every person in the building. The result is displayed as a privacy-respecting skeleton overlay ? no faces, no identifiable imagery.

This proposal asks Silent Nite Security to partner as the installation, monitoring, and maintenance provider for this system. I (Stefan) will develop and own the software. Silent Nite will handle deployment and operations. Together, we can offer something no other security company in the Free State currently can.

2. How It Works: WiFi Channel State Information

Every WiFi router transmits signals that travel through walls and bounce off objects ? including human bodies. Commercial WiFi chips already collect metadata called Channel State Information (CSI) to optimise their own signal quality. CSI captures how each subcarrier frequency is affected by amplitude changes and phase shifts caused by movement in the environment.

When a person walks, breathes, or even sits still in a room, their body reflects WiFi waves differently from empty space. These differences are measurable in the CSI data. A deep learning model (trained on thousands of examples) learns to map these signal patterns to body positions ? 17 keypoints including head, shoulders, elbows, hands, and knees.

The full pipeline:

- ? ESP32-S3 nodes capture raw CSI from ambient WiFi beacon frames (the router's regular signal)
- ? Signal processing removes noise and extracts motion patterns using wavelet transforms and spectral analysis
- ? A neural network reconstructs body pose ? 24 body regions tracked in real time
- ? Vital signs (breathing rate, heart rate) can also be extracted from the same signal
- ? Alert engine detects falls, loitering, perimeter breaches, or unexpected occupancy

This is not theoretical. Carnegie Mellon University published peer-reviewed results in 2018. Espressif (the ESP32 manufacturer) ships official CSI firmware under Apache 2.0 license. Open-source projects like WiFi-Radar and RuView provide working implementations with real-time dashboards, fall detection, and REST APIs.

3. New Capabilities for Silent Nite

This system solves problems that current alarm and CCTV systems cannot:

- ? **Through-wall detection:** The system detects people in rooms with no cameras. A locked store room, a barn, a neighbour's yard ? if WiFi reaches it, the system sees movement.
- ? **Works in total darkness:** WiFi signals are unaffected by lighting conditions. No IR, no night vision required.
- ? **Privacy-safe:** The output is a skeleton keypoint model (stick figure). No faces, no identifiable features. This addresses the growing objection homeowners have to indoor cameras.
- ? **CCTV integration:** The skeleton overlay can be drawn directly onto existing CCTV feeds. The camera shows what is visible; the WiFi shows what is behind the wall the camera cannot see.
- ? **Vital sign monitoring:** The same system detects if a person has stopped breathing or if their heart rate indicates distress. Useful for elderly care, child safety, and medical alerting.
- ? **Fall detection:** A sudden downward velocity exceeding a threshold triggers an immediate alert ? no wearable required.

4. Urban vs. Rural Deployment

This system works on any property with a WiFi network. Urban and suburban properties have advantages: smaller erf sizes mean fewer ESP32 nodes needed (typically 4-6), existing WiFi coverage is often excellent, and the density of properties creates overlapping coverage opportunities.

Rural and farm properties benefit most from the through-wall capability ? outbuildings, barns, and storage areas that cannot reasonably be wired for CCTV can be covered by a single ESP32 node within WiFi range of the main house. Farm watch patrols gain an additional detection layer for perimeter awareness.

Pricing scalability: a smaller urban property at around R100-R120/month is an accessible entry point. A large commercial or farm installation with 12+ nodes and dedicated monitoring could command R300-R500/month per site. This gives Silent Nite a tiered product line from budget to premium.

5. Proposed Partnership Model

Stefan (LAIS) ? Software & IP Owner

I will design, develop, and maintain the complete software stack: ESP32 firmware, central inference server, monitoring dashboard, alert engine, and CCTV integration layer. I retain copyright and will file a provisional patent on the CCTV-fusion method. Silent Nite receives exclusive deployment rights in the Free State for an initial period.

Silent Nite Security ? Operations & Deployment

Silent Nite's role would include:

- ? Procuring ESP32 hardware and server infrastructure (possibly through Staffix)
- ? Training installation technicians (1-day course provided by Stefan)
- ? Deploying and commissioning systems at client sites
- ? 24/7 monitoring through the existing control centre
- ? Ongoing maintenance, node replacement, and client support
- ? Client onboarding, billing, and relationship management
- ? Exploring custom hardware production via Staffix contacts ? a dedicated alarm eye beam designed specifically for CSI capture could dramatically improve range and reliability over generic ESP32 boards

Mutual Benefits

- ? Stefan gains a real-world deployment partner with an existing client base of 2000+
- ? Silent Nite gains a novel product offering no other Free State security company has
- ? Revenue sharing model aligns incentives ? both parties benefit from growth

6. System Design & Visual Feedback

The visual feedback software will run on a dedicated device provided by Silent Nite ? a small PC or NUC (Intel NUC or equivalent, approximately R3,000-R5,000) that sits in the control centre. This is kept separate from Silent Nite's existing alarm monitoring systems so there is zero risk of interference during the R&D phase.

System diagram:

- ? Each client site: 4-6 ESP32 nodes connected to the client's existing WiFi network
- ? ESP32s transmit CSI data over the internet to the Silent Nite central server
- ? Central server runs the ML inference engine and generates pose data, alerts, and events
- ? Dedicated monitoring workstation displays a live dashboard with property maps and skeleton overlays
- ? Provision CCTV streams can be ingested and shown side-by-side with the WiFi skeleton view
- ? Future: overlay WiFi skeletons directly onto Provision CCTV feeds on the same monitor

Provision cameras work well with this system because they are already IP-based and support RTSP streaming, which our integration layer is designed to consume. The existing Paradox alarm system provides the hard-wired security backbone ? our system adds a software sensing layer on top without modifying any alarm wiring or panel configuration.

7. Integration Roadmap

This is an R&D project. Realistic timeline:

- Phase 1 (Months 1-2):** Controlled environment testing. Deploy 4-6 ESP32 nodes at a Silent Nite site (office or a willing test client). Tune the ML model to local conditions. Validate detection accuracy against known ground truth (e.g., cameras + test walks).
- Phase 2 (Months 3-4):** Pilot deployment at 2-3 real client sites. Run alongside existing CCTV without replacing anything. Gather performance data. Refine alert thresholds. Train Silent Nite monitoring staff on the dashboard.
- Phase 3 (Months 5-6):** CCTV integration live ? skeleton overlay on Provision camera feeds. Begin client onboarding process development. Set up billing and support workflows.
- Phase 4 (Months 7-12):** Wider rollout based on pilot results. Develop tiered pricing. Explore Staffix custom hardware. Consider patent filing results.

8. Proposed Pricing Model

The entry-level pricing should be accessible to Silent Nite's existing urban residential clients. Current alarm monitoring sits at approximately R450/month. This system should be a low-barrier add-on, not a replacement.

Proposed Client Pricing (Monthly)

Tier	Nodes	Monthly	Best For
Starter	4	R99-R120	Small urban home
Standard	6	R130-R150	Suburban house + outbuildings
Premium	8-10	R200-R300	Large property, commercial
Enterprise	12+	R350-R500	Farm, warehouse, estate

The Starter tier at R99-R120/month is intentionally low ? it matches the goal of being under R150 and acts as an entry point. For Silent Nite's existing 2,000+ clients, even a 5% conversion rate (100 clients) at the Starter tier would generate R9,900-R12,000/month in new recurring revenue.

If 200 existing clients convert at mixed tiers (Starter, Standard, Premium), the monthly recurring revenue from this product alone could reach R35,000-R50,000/month ? all from existing relationships. New clients gained through this unique offering would be additional.

9. Client Onboarding

For existing clients with Paradox alarm systems and Provision CCTV:

- ? Onsite survey ? 30 minutes to map the property and plan node placement (done by Silent Nite technician)
- ? Node installation ? 1-2 hours for a standard property. Mount ESP32s, connect to existing WiFi
- ? Commissioning ? technician confirms all nodes are reporting, runs a walk test
- ? Client app access ? homeowner can view their own property status (optional)
- ? Monitoring handover ? system goes live in Silent Nite's control centre alongside their existing alarms

10. What Stefan Needs From Silent Nite

To proceed with development and testing, I require:

- ? A test site ? a Silent Nite office location or a willing client property where we can deploy a pilot system
- ? Access to a Provision CCTV stream for integration testing (the dedicated NUC will read this and overlay pose data)
- ? Contact with Staffix to discuss custom ESP32 hardware possibilities ? a purpose-built alarm eye beam form factor would be ideal
- ? Commitment to a 3-month trial period with defined success metrics (detection accuracy, false alarm rate, client satisfaction)
- ? Designation of a technical contact at Silent Nite who will coordinate installation and provide feedback

11. Risk & Limitations

Honest assessment of current limitations:

- ? ESP32 reliability is not yet alarm-grade ? early builds may need occasional resets or reflashing
- ? Dependent on client WiFi ? if the network goes down, the sensing layer goes offline (existing alarm systems have cellular backup; this does not)
- ? Range is limited to approximately 20-30ft through drywall ? concrete or metal structures reduce this
- ? The ML model benefits from per-environment calibration ? installation technicians need training
- ? This is an R&D project. The first 6 months will involve iteration and refinement, not a finished product

These limitations are manageable. The technology is proven in research and open-source implementations. What does not exist yet is a production-grade, security-industry-ready product. That is what this partnership would build.

12. Next Steps

To move forward, I propose:

- ? A meeting between Stefan and Luigi to discuss this proposal in person
- ? A site visit to Silent Nite's control centre to understand current infrastructure
- ? Agreement on a test site for Phase 1 deployment
- ? Non-disclosure agreement if required
- ? Signature of a simple R&D partnership letter of intent

Contact

Stefan ? LAIS Local AI System

Email: lais.localaisystem@gmail.com

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The WiFi sensing technology described builds upon prior academic research and open-source implementations (CMU RF-Pose, Espressif ESP-CSI, WiFi-Radar) and does not claim to patent Channel State Information sensing itself. Patent protection may be sought for novel integration methods (CCTV-fusion overlay system) described herein.

No binding agreement is formed by this document. A separate contract would govern any joint development activities.